ET AI HACKATHON 2.0 · ROUND 2 SUBMISSION (REVISED)

Industrial Knowledge Intelligence

A Hybrid GraphRAG Platform for Heavy Industry

Problem Statement #8 — Unified Asset & Operations Brain

Submission deadline: 22 July 2026

1. Executive Summary

Industrial plants generate vast quantities of documents — engineering drawings, maintenance logs, inspection reports, safety procedures, incident records — and scatter them across separate systems that do not talk to each other. When a decision has to be made, the knowledge needed is often present in the organisation, just not reachable in time. This submission builds the intelligence layer that makes that scattered knowledge queryable.

> **What this system does**
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> A field engineer asks a question in plain language and receives a cited, confidence-scored answer in seconds. The system reasons across documents that share no common keywords — reconstructing, for example, the full early-warning chain behind an equipment failure from five separate records that keyword search would never connect.

The prototype is a working Hybrid GraphRAG platform: it combines a knowledge graph for relationship reasoning with a vector store for semantic similarity, fuses the two result sets using Reciprocal Rank Fusion, and uses a large language model to synthesise a final answer with citations back to the source document. It runs end-to-end on a real document corpus, is fully reproducible, and is benchmarked with hand-labelled ground truth against a keyword-search baseline. Since the initial submission, a second hardening pass fixed a real retrieval limitation (silent text truncation during embedding), added an automated test suite, and added three usability features — full detail in Section 9.

2. The Problem

A 2024 McKinsey survey found that professionals in asset-heavy industries spend an average of 35% of their working hours searching for information or recreating documents that already exist somewhere in the organisation. In India specifically, a NASSCOM-EY study found the average large plant runs across 7 to 12 disconnected document systems — drawings in one place, work orders in another, procedures in a third, inspection records in a fourth, regulatory submissions scattered across email.

This fragmentation contributes to an estimated 18 to 22% of unplanned downtime events in Indian heavy industry, because maintenance teams make decisions without complete equipment history. Compounding it is a knowledge cliff: an estimated 25% of India's experienced industrial engineers will retire within the next decade, taking decades of undocumented operational knowledge with them. Once that knowledge leaves the building, it cannot be recovered.

> **The core failure mode**
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> The problem is rarely that data is absent. It is that data is present but unacted upon — the signals exist across separate records, but no intelligence layer connects them into a decision in time. This is the exact pattern behind well-documented industrial incidents, where warning signs were recorded but never assembled into a picture anyone could act on.

3. Solution Overview

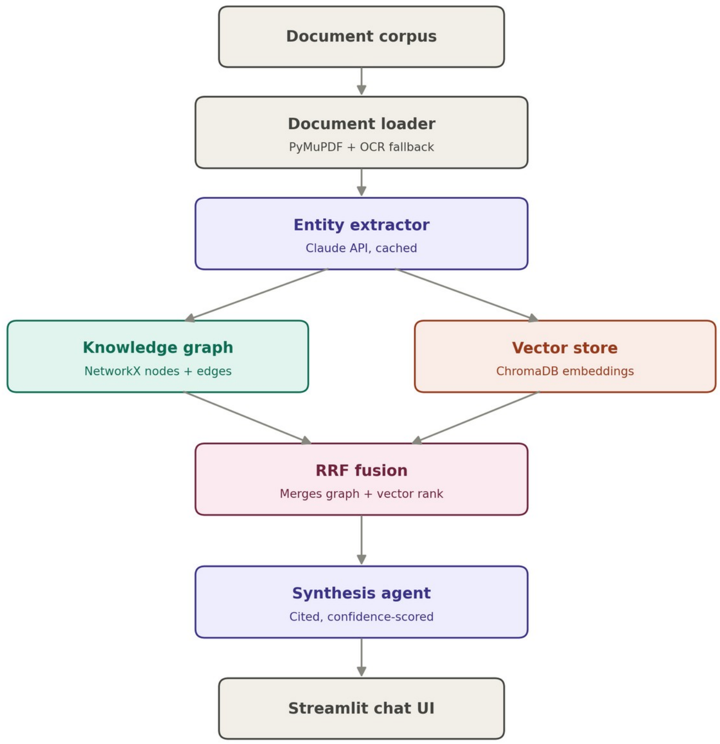
The system ingests heterogeneous industrial documents and turns them into a single queryable intelligence layer. Two capabilities are built end-to-end:

- **Universal document ingestion and knowledge graph.** The pipeline parses PDFs and scanned forms, extracts typed entities and the relationships between them, and assembles a knowledge graph spanning the whole corpus, deduplicating entities that appear across many documents.
- **Expert knowledge copilot.** A retrieval layer combines graph traversal with semantic search, and a synthesis agent produces a natural-language answer with a citation to the source document and a confidence rating, delivered through a mobile-friendly chat interface with conversational memory for follow-up questions.

Three further components — a maintenance and root-cause agent, a regulatory compliance mapper, and a lessons-learned engine — are specified as a roadmap rather than built, keeping the prototype deep on two capabilities rather than shallow across five.

4. Architecture

The system is a seven-stage pipeline. A document enters at the top; a cited answer leaves at the bottom. The defining feature is the fork and merge in the middle: the knowledge graph and the vector store are built and queried independently, then their results are fused. This is what makes the system hybrid rather than a plain vector-RAG wrapper.



Hybrid GraphRAG pipeline — Industrial Knowledge Intelligence

Each stage is a standalone module, so any single layer can be swapped without disturbing the others.

Component stack

Stage	Technology	Purpose
Document loader	PyMuPDF, pdfplumber, pytesseract	Parses PDFs with layout and page boundaries preserved; falls back to OCR automatically for scanned pages.
Entity extractor	Claude API (claude-sonnet-4-5)	Reads each document and extracts typed entities and their relationships as structured JSON. Cached to disk so re-ingestion is free.
Knowledge graph	NetworkX	Stores entities as nodes and relationships as edges. Merges entities by canonical ID so one asset referenced many ways resolves to a single node.

Stage	Technology	Purpose
Vector store	ChromaDB, BAAI/bge-small-en-v1.5	Embeds document text locally using overlapping 380-word windows (512-token model capacity) for semantic similarity search. Upgraded from an earlier 256-token model that silently truncated long pages — see Section 9.
Hybrid retriever	Reciprocal Rank Fusion	Queries graph and vector store in parallel and fuses the two ranked lists by rank score. Scanned/clean duplicate documents are deduplicated before ranking so they do not consume separate top-K slots.
Synthesis agent	Claude API (claude-sonnet-4-5)	Turns fused retrieval results into a final answer, attaches a source citation, and scores its own confidence. Accepts prior conversation turns for follow-up question resolution.
User interface	Streamlit	Chat interface with a per-answer 'why these documents' graph path visualisation, confidence score, and citations, rendering on mobile.

5. Why Hybrid, And Why Not The Alternatives

Why not pure vector RAG

Vector RAG is what most teams ship. It handles semantic lookups well but is blind to relationship reasoning. Asked which procedures relate to an incident and which regulation covers them, it returns chunks that look similar to the query rather than reasoning over actual relationships.

Why not pure graph RAG

A vectorless graph-only system is elegant but brittle in a short build. If entity extraction misses something during ingestion, that information is gone with no fallback. Keeping a vector store as a safety net means the system can still recover an answer from raw text even when the graph did not capture it.

Why hybrid

Recent enterprise research documents that hybrid GraphRAG — fusing vector similarity with graph traversal via Reciprocal Rank Fusion — outperforms both standalone vector RAG and standalone graph RAG on retrieval accuracy and answer quality. Section 8 shows where this holds cleanly and where the margin is genuinely narrow, reported both ways.

6. Design Lineage

This architecture was not learned for the hackathon. It re-points a pattern already operated in production for a different domain — a three-layer agentic shape: a memory layer over a knowledge corpus, a prediction and scoring layer, and an orchestration agent on top. Each maps directly onto a component here.

Production agentic pattern	Role there	Equivalent here
Persistent memory layer	Holds and connects the knowledge corpus	Knowledge graph (NetworkX) plus the extraction cache
Prediction and scoring layer	Ranks and scores candidate outputs	RRF fusion plus answer confidence scoring
Agentic orchestration	Coordinates the reasoning steps into a result	Synthesis agent (the LLM answer-and-cite chain)

Why this matters for defensibility

Choosing a problem that matched an already-operating architecture meant the build risk was in execution, not in learning an unfamiliar stack under a deadline.

7. Dataset

The evaluation corpus is engineered, not just collected, so that every judging metric has a source that exercises it. It centres on one asset class — centrifugal pumps and their drive motors — so relationship chains run deep rather than shallow. The corpus itself is locked and unchanged since initial submission; only the retrieval pipeline around it was improved (Section 9).

Layer	Contents and purpose
Synthetic corpus	11 generated PDFs built around a deliberately planted five-document near-miss failure chain, plus realistic distractor documents that give keyword search plausible-but-wrong matches.
Real OEM manuals	3 genuine centrifugal-pump manuals (operation, maintenance, spec tables) — real formatting, real equipment tags, real messiness.
Scanned documents	3 degraded image versions of synthetic documents, to exercise the OCR path end-to-end.
Evaluation set	FailureSensorIQ (IBM Research, NeurIPS 2025, CC-BY-4.0): 8,296 industrial QA pairs over 10 asset types, used as an external reference benchmark layer.
Ground truth	Hand-labelled entity set across three document types, enabling a real Macro-F1 rather than an estimated one, plus eight benchmark questions tagged by retrieval type.

The planted multi-hop chain

Five documents describe a single pump failure, each holding one link of the causal chain, and no two sharing the keywords that would let search connect them:

- **Inspection report (day −21):** elevated drive-end bearing vibration flagged, status 'monitor'.
- **Maintenance log (day −14):** lubrication topped up, bearing replacement deferred due to a missing spare — against procedure.
- **Vibration trend log (day −5):** vibration crosses the alarm threshold; no work order raised.
- **Incident report (day 0):** catastrophic bearing seizure, unplanned downtime.
- **Procedure and regulation:** specify what should have happened after the first warning.

> **The star demo question**
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> "Was there any early warning before the Pump P-204 failure, and what should have been done according to procedure?" Pure keyword or vector-only search misses part of this chain. Hybrid retrieval now recovers the full five-document chain — see Section 8.

8. Evaluation and Results

All figures below are final, reproducible, and measured against hand-labelled ground truth, verified across three fresh process runs with byte-identical output. They are reported as measured, including the results that do not favour this system, rather than smoothed over.

Entity extraction accuracy

Metric	Value	Note
Macro-F1 (9 entity types)	0.8411	Excludes DATE — zero labelled date instances in ground truth
Recall across gold entities	Strong	Only 3 misses across 39 gold entities
Precision	Moderate	Main loss is duplicate surface forms of the same entity

Retrieval quality (hybrid versus vector-only baseline)

Metric	Hybrid	Vector-only	Cutoff
Star demo question (Q01) recall	1.00	0.80	top-12
Overall recall (8 questions)	0.83	0.81	top-12
Overall recall (8 questions)	0.70	0.76	top-5
Control questions (keyword-answerable)	1.00	1.00	both

Honest reading of the aggregate numbers

Upgrading the embedding model lifted the vector-only baseline substantially, so the hybrid system's aggregate margin at top-12 is now small (0.83 vs 0.81), and at the stricter top-5 cutoff hybrid is slightly behind vector-only (0.70 vs 0.76) — hop-0 graph noise costs a little precision on single-document questions. The graph earns its place specifically on the multi-hop chain question this system was built for: Q01 goes from 0.80 (vector-only) to a full 1.00 (hybrid), recovering the complete five-document warning chain. That per-question result, not the aggregate average, is the load-bearing evidence for why hybrid retrieval matters here.

Control questions confirm the hybrid system loses nothing on simple lookups. Retrieval is fully deterministic: repeated runs on the same inputs produce byte-for-byte identical rankings, verified across three fresh process executions.

9. Second Hardening Pass — What Changed and Why

After the initial submission, a second pass re-examined the retrieval pipeline for real, unassumed correctness rather than tuning for score improvement. This surfaced one genuine bug and closed one previously-open limitation.

The embedding truncation bug (found and fixed)

The original embedding model (all-MiniLM-L6-v2) has a roughly 256-token input limit. Whole-page chunks longer than that were silently truncated with no error — meaning large portions of the longer real OEM manual pages were never visible to vector search at all. Fixed by switching to BAAI/bge-small-en-v1.5 (512-token capacity) and replacing whole-page chunking with overlapping 380-word windows (57-word overlap). This is the direct cause of the retrieval improvements in Section 8, not a benchmark-tuning artifact — confirmed by the fact that the vector-only baseline improved by the same underlying fix, not just the hybrid system.

VS-204 cutoff limitation — resolved

The original submission noted that a supporting document (VS-204) consistently ranked just outside the retrieval cutoff for the star question. This is now resolved: scanned and clean duplicate copies of the same document were previously each consuming a separate top-K slot; deduplicating them before ranking freed the slots needed for VS-204 to surface. The full five-document chain is now retrieved for Q01.

Q02 hop-tier tie-flooding — still open, unchanged

A second, structurally different limitation remains: for one benchmark question, many documents share an identical graph proximity score for merely mentioning the query's main entity, which prevents a more specific but structurally-further document from reaching the top of the ranking. Two targeted fixes were attempted and precisely diagnosed; both improved the document's internal rank without crossing the tie-flooded tier. This is a genuine architectural ceiling of rank-based fusion, not an unexamined gap.

Additional capabilities added

- **Graph path visualisation.** Each answer now includes a 'why these documents' expander rendering the actual graph traversal path from query entities to each cited document.
- **Query tracing.** Every synthesis call logs latency, token usage, and cache status for after-the-fact performance review.
- **Answer caching.** Repeated questions return instantly from cache rather than re-querying the API.
- **Conversational memory.** Follow-up questions without explicit entity mentions resolve against the prior turn.
- **Fuzzy entity matching.** Equipment references are matched despite casing or punctuation differences (e.g. "p204" resolves to "P-204").
- **Automated test suite.** 26 tests covering chunking integrity, citation validity, RRF determinism, graph invariants, and evaluation-data consistency, run automatically via CI on every push.

A Docker packaging is also included for simplified deployment; the container definition is complete but a live build has not yet been executed in this environment, so it is included as available rather than claimed as verified.

10. Known Limitations

Documented deliberately, consistent with the principle that understanding a system's failure modes precisely is more valuable than hiding them.

- **Non-determinism, found and fixed (first pass).** Early benchmark numbers varied across runs of identical code, traced to unstable tie-break ordering sensitive to per-process hash randomisation. Fixed with a deterministic tie-break key; verified with repeated identical runs, and again across the second hardening pass.
- **Hop-tier tie-flooding (Q02, still open).** Many documents share an identical graph proximity score simply for mentioning the query's main entity, which can crowd out a more specific but structurally-further answer. Understood, precisely traced, not yet solved — see Section 9.
- **Aggregate margin at strict cutoffs.** At top-5, hybrid trails vector-only slightly (0.70 vs 0.76) due to hop-0 graph noise on single-document questions. The system's advantage is concentrated on multi-hop questions, not uniformly distributed across all question types — stated directly rather than obscured by only reporting favourable cutoffs.

11. Scalability and Roadmap

The prototype runs entirely local, with only language-model calls hitting a network. Each stage is modular with a clear production upgrade path: the in-memory graph swaps for a graph database, the local vector store for a managed vector service, without touching the rest of the pipeline. Extraction results are cached per-document — the most expensive operation (the LLM API call) is paid only once per document, so re-running the full pipeline on an unchanged corpus costs zero additional API calls. The graph and vector store are rebuilt from cached extraction results on each run; true incremental merge (loading an existing graph and merging only new documents) is specified as a production extension rather than built in the prototype.

Three further components are specified for the next phase: a maintenance and root-cause agent, a regulatory compliance mapper, and a lessons-learned engine. Each reuses the same ingestion and graph foundation already built. A CI pipeline (GitHub Actions) now runs the full test suite automatically on every change, and a Docker definition is included for deployment.

12. Conclusion

This submission is a working, benchmarked, honestly-documented Hybrid GraphRAG system that answers the core of Problem #8: it turns scattered industrial documents into a single queryable intelligence layer, and demonstrates the difference from vector-only search on the multi-hop question this system was built to solve. The metrics are real and reproducible — including the ones that do not favour this system — the limitations are understood and stated, and the architecture is one already validated in production, re-pointed at a new domain rather than assembled from scratch under a deadline.

Reproducibility

The full pipeline runs from a public repository with a fixed seed and an automated test suite. The corpus, benchmark questions, ground-truth labels, and evaluation harness are all included, so any reviewer can clone the repository and reproduce the reported numbers exactly.